

Course Code: CS1211

Name of the Programme: B.Tech(H)

Name of the Course: Database Management Systems

Semester:2

Course Credit	No. of hours per week (L + T + P)	Total no. of Teaching hours
4	30+15+30	75

Course Objectives:

- A. To describe fundamental concepts of database systems, including data models, architectures, and schema design.
- B. To analyse and decompose database designs using normalisation and functional dependencies to eliminate redundancy and improve efficiency
- C. To solve real-world problems by applying the Database Management System concept.
- D. Learn the necessary skills for transaction processing and effective database design.

Syllabus:

Module – 1 Introduction to Database Management Systems	No. of hours
	6
Database system Vs file system, Database system architecture, Data model schema and instances, Data independence and Database language and interfaces - DDL, DML, DCL and TCL Overall Database Structure - Storage Components, Metadata, Users Database Backups and Recovery Mechanisms.	

Module – 2 Entity Relationship Modelling and Constraints in Database	No. of hours
	6
ER model concepts, notation for ER diagrams, mapping constraints, reduction of ER diagrams to tables, relationship of higher degree Integrity constraints, entity integrity, referential integrity, Keys constraints, Domain constraints, relational algebra and relational calculus.	

Module – 3 SQL Queries	No. of hours
	4
Characteristics of SQL, advantages of SQL. SQL data type and literals, Types of SQL commands, SQL operators and their procedure, Tables, views and indexes, Queries and subqueries, Aggregate functions. Insert, update and delete operations, Joins, Unions, Intersection, Minus, Triggers and Procedures in SQL.	

Module – 4 Normalization	No. of hours
	7
Functional dependencies, normal forms, first, second, third normal forms, BCNF, inclusion dependence, lossless join decompositions, normalization using FD.	

Module – 5 Transaction Processing and Concurrency Control Concept	No. of hours
	7
Transaction, ACID Properties, testing of serializability, serializability of schedules, conflict & view serializable schedule, Deadlocks, Locking Techniques for concurrency control, Time stamping protocols for concurrency control.	

Course Outcomes
a) Explore the database concepts, architecture and transaction for DBMS.
b) Design a database application to solve real world problems using ER Diagram.
c) Identify and solve the redundancy problem in database tables using normalization.
d) Apply transaction concepts and concurrency control for data integrity and performance in DBMS.

Text Books	
1	A. Silberschatz, H. F. Korth, and S. Sudarshan, <i>Database System Concepts</i> , 7th ed. New York, NY, USA: McGraw-Hill, 2019, ISBN: 978-9390727506.
2	R. Elmasri and S. B. Navathe, <i>Fundamentals of Database Systems</i> , 7th ed. New Delhi, India: Pearson, 2016, ISBN: 978-9332582705.

Reference Books	
1	R. Ramakrishnan and J. Gehrke, <i>Database Management Systems</i> , 3rd ed. New York, NY, USA: McGraw-Hill, 2002, ISBN: 978-0072465631.

Lab Programs	
1.	Implementation of DDL Commands in SQL: Demonstrate the use of DDL commands with suitable examples, including CREATE TABLE, ALTER TABLE, and DROP TABLE.
2.	Implementation of DML Commands in SQL: Implement DML commands with suitable examples, including INSERT, UPDATE, and DELETE.
3.	Implementation of SQL Functions: Implement various types of functions in SQL with examples, including Number Functions, Aggregate Functions, Character Functions, Conversion Functions, and Date Functions.
4.	Implementation of SQL Operators: Demonstrate the use of different types of operators in SQL with examples, including Arithmetic Operators, Logical Operators, Comparison Operators, Special Operators, and Set Operations.
5.	Implementation of Different Types of Joins: Implement various types of joins in SQL, including Inner Join, Outer Join, and Natural Join, with suitable examples.
6.	Study and Implementation of Grouping and Ordering: Study and implement the GROUP BY and HAVING clauses, ORDER BY clause, and Indexing in SQL.
7.	Backup and Recovery for Data Safety: Implement backup and recovery commands in SQL to ensure data safety.
8.	Managing Users and Administering Database Tasks: Implement tasks for managing database users and administering database operations.

Tutorials	
1	Database Architecture & Components
2	ER Model & Diagram Representation
3	Converting ER Diagrams to Tables
4	Integrity Constraints & Keys
5	Introduction to SQL & Basic Queries
6	DML & DDL Commands
7	Joins & Subqueries
8	Functional Dependencies & 1NF, 2NF and 3 NF